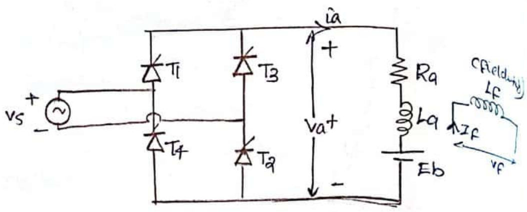

KTU
NOTES

dc motor drives

Imp — Singlephase fullycontrolled rectifier fed a separately excited dc motor.

The drive ckt is shown in fig(a)

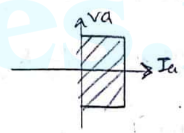


Motor is shown by its equilent ckt. The input Ac voltage $V_s = V_m \sin \omega t$

field winding is excited by a separate dc source

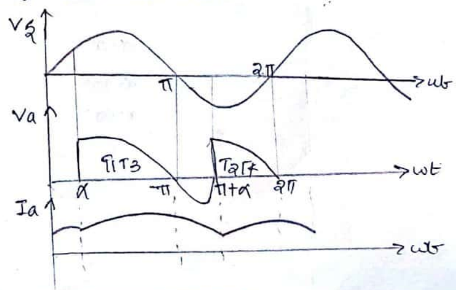
Continuous armature current

In the full converter s/m, thyristors T_1 and T_3 are simultaneously triggered at α and thyristors T_2 and T_4 are triggered at $\pi+\alpha$. Here the motor is always connected to the input supply through thyristors. T_1 and T_3 conduct during the interval $\alpha < \omega t < \pi+\alpha$ and connect the motor to the supply. At $\pi+\alpha$, T_2 and T_4 are triggered, immediately the supply voltage applied across thyristors T_1 and T_3 are as reverse bias voltage and turns them off. This is called natural commutation. The motor current I_a which was flowing from the supply through T_1 and T_3 is transferred to T_2 and T_4. During α to π energy flows from input supply to the motor signifying positive powerflow. ie, V_a and I_a are +ve. However during π to $\pi+\alpha$. Some of the motor s/m energy is fed back to the i/p system. ie, V_a is −ve, I_a is +ve signifying reverse powerflow.



This is also called 2 quadrant converter.

The voltage and current waveforms under continuous conduction are shown below



Torque-speed characteristics

The avg value of o/p voltage for 1φ fullconverter in the armature ckt $V_a = \frac{2V_m}{\pi}\cos\alpha$.

$$N \propto E_b/\phi$$

that is, $\omega_m \propto \frac{E_b}{\phi}$

Flux is constant.

$$\omega_m = k_e E_b$$

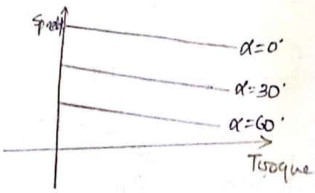
$$\omega_m = k_e(V_a - I_a R_a)$$

also $T \propto \phi I_a$

$$T = k_e \phi I_a \quad \text{(since } \phi \text{ is constant)}$$

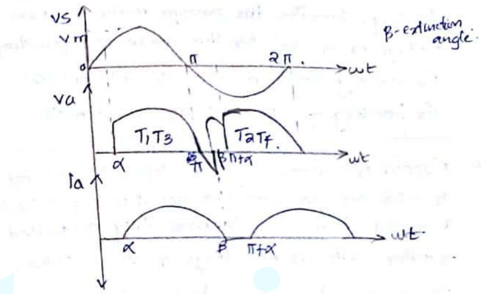
$$\therefore \omega_m = k_e\left(\frac{2V_m}{\pi}\cos\alpha - \frac{T}{k_e}R_a\right)$$

$$\omega_m = \frac{k_e 2V_m}{\pi}\cos\alpha - TR_a$$



Discontinuous armature current

The voltage and current waveforms for fullconverter with discontinuous armature current are shown in the figures below



SCRs T_1 and T_3 are fired at α and armature current flows from α to β. The motor armature is connected to the supply from α to β. From β to π+α, the motor costs (slowdown) and the armature voltage is the backemf of the motor. Again at π+α, SCRs T_2 and T_4 are triggered, and the condition conduction continues upto π+β.

β is called extinction angle.

Note:- when the machine is acting as a generator and ... back the power to the ac supply. This is ... as inversion operation of the converter. and this mode of operation of is used as regenerative breaking. Therefore the average motor terminal voltage Va is -Ve for this mode of operation.

for motoring operation, $\alpha < 90^\circ$ and $\omega_m > 0$.

for breaking operation, $\alpha > 90^\circ$ and $\omega_m < 0$

Imp ? A 200V, 875 rpm, 150 Amps, Separately excited dc motor has an armature resistance of 0.06 Ω. It is fed from a single phase fully controlled rectifier with an AC same voltage of 220V, 50Hz assuming continuous conduction. Calculate

i) Firing angle for rated motor torque and 750 rpm

(H.W) ii) Firing angle for rated motor torque and -500 rpm

iii) Motor speed for $\alpha = 160^\circ$ and rated torque (10 mark)

Answer :

$V_s = 220V$ (rms)

$f = 50Hz$.

$V_a = 200V$

$N = 875 rpm$

$I_a = 150$ rpm

i) $V_{a1} = 200V$

$V_{a2} = ?$

$N_1 = 875 rpm$.

$I_{a1} = 150 Amp$.

$R_a = 0.06 \Omega$.

$E_{b1} = V_{a1} - I_{a1}R_a$

$= 200 - (150 \times 0.06) = \underline{191V}$

for 750 rpm ie, $N_2 = 750 rpm$.

$N \propto \frac{E_b}{\phi}$

$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$ ($\because$ flux is constant)

$\frac{750}{875} = \frac{E_{b2}}{191}$

$E_{b2} = \underline{\underline{163.71V}}$

$E_{b2} = V_{a2} - I_{a2}R_a$. Here. ($I_{a1} = I_{a2} \Rightarrow$ rated motor torque)

$V_{a2} = 163.71 + (150 \times 0.06)$

$= \underline{\underline{172.71 V}}$

$T \alpha \phi I_a$

$T \alpha I_a$.

$I_{a1} = I_{a2}$

$\therefore V_{a2} = \frac{2V_m}{\pi} \cos\alpha$

$\cancel{V_m = \frac{V_s}{\sqrt{2}} = \frac{220}{\sqrt{2}} = 155.56}$.

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$V_m = \sqrt{2} \times 220 = \underline{311.12\,V}$$

$$172.71 = \frac{2 \times 311.12}{\pi} \cos\alpha$$

$$\cos\alpha = 0.871$$

$$\underline{\alpha = 29.3^\circ}$$

iii) V_a $\alpha = 160^\circ$

$$V_{a2} = \frac{2V_m}{\pi} \cos\alpha.$$

$$= \frac{2 \times 311.12}{\pi} \times \cos(160)$$

$$= \underline{-186.12\,V}$$

$$V_{a2} = E_{b2} + I_{a2} R_a$$

$$E_{b2} = -186.12 - (150 \times 0.06)$$

$$= \underline{-195.12\,V}$$

$$\frac{N_2}{875} = \frac{-195.12}{191}$$

$$\underline{N_2 = -893.87\ rpm}$$

ii) $\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$

$$\frac{-500}{875} = \frac{E_{b2}}{191} \Rightarrow E_{b2} = \underline{-109.14\,V}$$

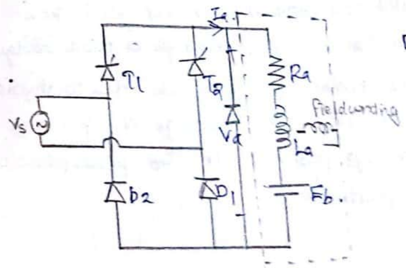
$$V_{a2} = -109.14 + (150 \times 0.06) = \underline{-100.14}\ V$$

$$V_{a2} = \frac{2V_m}{\pi} \cos\alpha$$

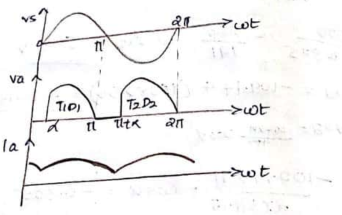
$$\frac{-100.14 \times \pi}{2 \times 311.12} = \cos\alpha = -0.505$$

$$\underline{\alpha = 120.33^\circ}.$$

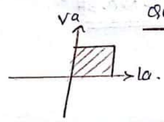
Single phase semicontrolled Rectifier fed a separately excited dc motor. (10 mark)



★ Continuous conduction



Quadrant of operation



Semiconvertors are one quadrant converters that is they have one polarity of voltage and current at the dc terminals. The circuit arrangement for singlephase semiconverter drive is shown in fig above. The armature voltage is always controlled by a semiconverter and field ckt is fed from the AC supply through a diode bridge. The motor current cannot reverse due to thyristors in the converters. The average dc o/p voltage V_a is always positive. Therefore power flow is always positive.

Continuous armature current.

The voltage and current waveforms are shown above. SCR T_1 is triggered at an angle 'α' and T_2 at an angle $\pi+\alpha$, with respect to the supply voltage V_s. For the period from α to π, the motor is connected to the input supply through T_1 and D_1 and the motor terminal voltage V_a is same as the supply input voltage. beyond π, V_a tends to reverse as the input voltage changes polarity. But with the forward bias the freewheeling diode D_F the motor terminals are shortcircuited through D_F making V_a as zero. Thus the motor current I_a which was flowing from the supply through T_1 is transferred to D_F.

When thyristor conducts from period α to π, energy from the supply is delivered to the armature ckt. This energy is partially stored in inductance, partially stored in the K.E of the moving s/m and partially used to supply the mechanical load. During the freewheeling period energy is recovered from the inductance and it is converted into mechanical force to supplement the K.E in supplying the mechanical load. During this period no energy is feedback to the supply.

Torque speed characteristics.

Average value of O/P voltage

$$V_a = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin\omega t \, d\omega t$$

$$= \frac{V_m}{\pi}\left[-\cos\omega t\right]_{\alpha}^{\pi}$$

$$= \frac{V_m}{\pi}(1+\cos\alpha)$$

we have speed N or $\omega_m \propto E_b/\phi$

$N \propto \frac{E_b}{\phi}$ {ϕ is constant}

$E_b = kN$

$= k\omega_m$

$\therefore \omega_m = \frac{E_b}{k}$

$= \frac{V_a - I_a R_a}{k}$

$T \propto \phi I_a$

also $T \propto I_a$ (ϕ constant)

$T = k I_a$.

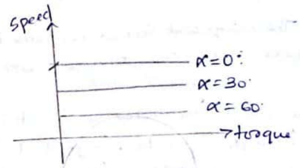
Thus speed

$$\omega_m = \frac{V_m}{\pi}(1+\cos\alpha) - \frac{T R_a}{K^2}$$

At no load $T=0$, ie, $\omega_{mo} = \frac{V_m}{\pi K}(1+\cos\alpha)$

$\alpha \uparrow$ speed $\downarrow$

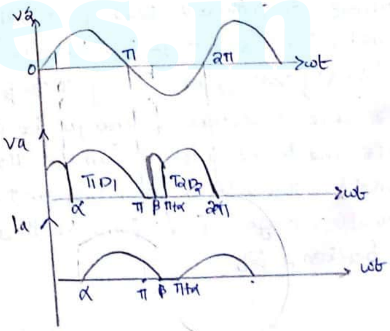
torque-speed chara.



As $\alpha \uparrow$ speed $\downarrow$

⇒ Discontinuous Armature Current

Back emf developed due to DF



The armature current become discontinuous for large values of firing angle, high speed and low values of torque. The motor performance diminishes with discontinuous armature current. Therefore it is desirable to operate in continuous current mode.

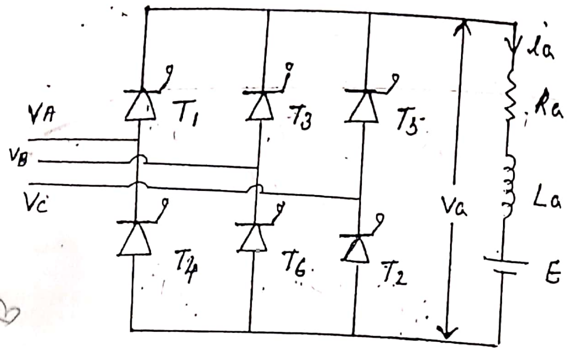
The voltage and current waveforms for a 1ϕ semiconverter with discontinuous current are shown in fig above.

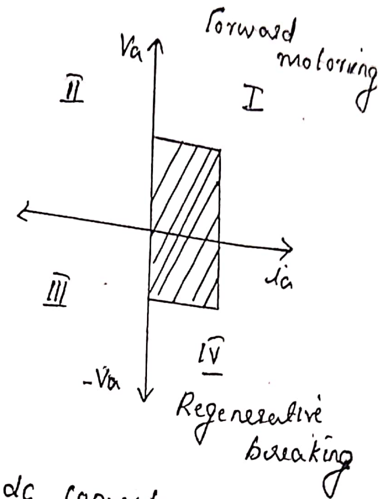
For the period α to π the motor is connected through the supply through T_1 and D_1. Beyond π the motor terminal is shortcircuited through D_F. The armature current decays to zero at an angle β (extension angle) before the thyristor T_2 is triggered at $\pi+\alpha$, thereby making armature current discontinuous. During α to π, the motor terminal voltage V_a is same as that of supply voltage V_S. During π to β the motor current freewheels through D_F and so V_a is zero. In the interval from β to $\pi+\alpha$, the motor slows down and the motor terminal voltage V_a is same as that of the back emf E_b.

THREE PHASE FULLY-CONTROLLED DC DRIVE FOR SEPERATELY EXCITED DC MOTOR

Nirmala Paul (1)

M-II





For large power dc loads, 3ϕ ac to dc converters commonly used. The various types of three-phase phase controlled converters are 3phase half-wave converter, 3-phase semiconverter, 3phase full converter and 3phase dual converter. Three phase half-wave converter is rarely used because it introduces dc component in the supply current.

Fig shows threephase fully-controlled converter with RLE as load. is supplied by three phase supply ABC.

$$V_A = V_m \sin \omega t$$

$$V_B = V_m \sin(\omega t - 120)$$

$$V_C = V_m \sin(\omega t + 120)$$

Three phase full converter is preferred where the regeneration is required. Thyristors are fired in the sequence they are numbered with a phase difference of 60°. The line commutation of an odd numbered thyristor occurs with the turning on of the next odd numbered thyristor. The same is true for

The same is true for even numbered thyristors consequently, each thyristor conducts for 120° and only thyristors conduct at a time. one odd numbered and one even numbered.

The transfer of current from an outgoing to incoming thyristor can take place when the respective line voltage is of such a polarity that not only does it forward bias the incoming thyristor but it also leads to reverse biasing of the outgoing thyristor. when the incoming thyristor is turned ON.

when The thyristors are conducting in the following seques and the o/p voltage corresponding to each of them are given below

T_1T_2, T_2T_3, T_3T_4, T_4T_5, T_5T_6, T_6,T_1 - - - - -

$T_1\ T_2 \rightarrow V_{ac}$

$T_2\ T_3 \rightarrow V_{bc}$

$T_3\ T_4 \rightarrow V_{ba}$

$T_4\ T_5 \rightarrow V_{ca}$

$T_5\ T_6 \rightarrow V_{cb}$

$T_6\ T_1 \rightarrow V_{ab}$

$T_1\ T_2 \rightarrow V_{ac}$.

line voltage corresponding to thyristor sequence is

V_{ac}, V_{bc}, V_{ba}, V_{ca}, V_{cb}, V_{ab}, V_{ac} - - - -

The average o/p voltage will be line voltages.

In three phase full converter each thyristor pair will conducts for 60° & each thyristor will conducts for 120°. For example T_1 conductor for 120° 60° with T_6 & 60° with T_2. with T_6, T_1 will give the line voltage Vab & with ... it will give the line voltage Vac

when $\alpha = 0°$

For $\alpha = 0°$ the thyristors will behave like diodes. For $\alpha = 0°$, T_1 is trigged at $\omega t = \pi/6$ (the most positive voltage and T_2 is trigged at $\omega t = \pi/2$ at the most negative voltage corresponding to the phase voltage. Since line voltage leads phase voltage by 30°, while considering the line voltages T_1 is triggered at $\omega t = 60°$ ($\pi/3$).

$\omega t = \alpha + 30°$

Each SCR conducts for 120°, when T_1 is trigged, reverse biased T_5 is turned off and T_1 is turned ON. T_6 is already conducting. As T_1 is connected to A and T_6 connected to B voltage V_{AB} appears across the load. When T_2 is turned ON T_6 is commutated. T_2 is from the negative group T_6 is commutated. As T_1 and T_2 is conducting T_1 is already conducting. As T_1 and T_2 is conducting the line voltage appears across the load is V_{AC}.

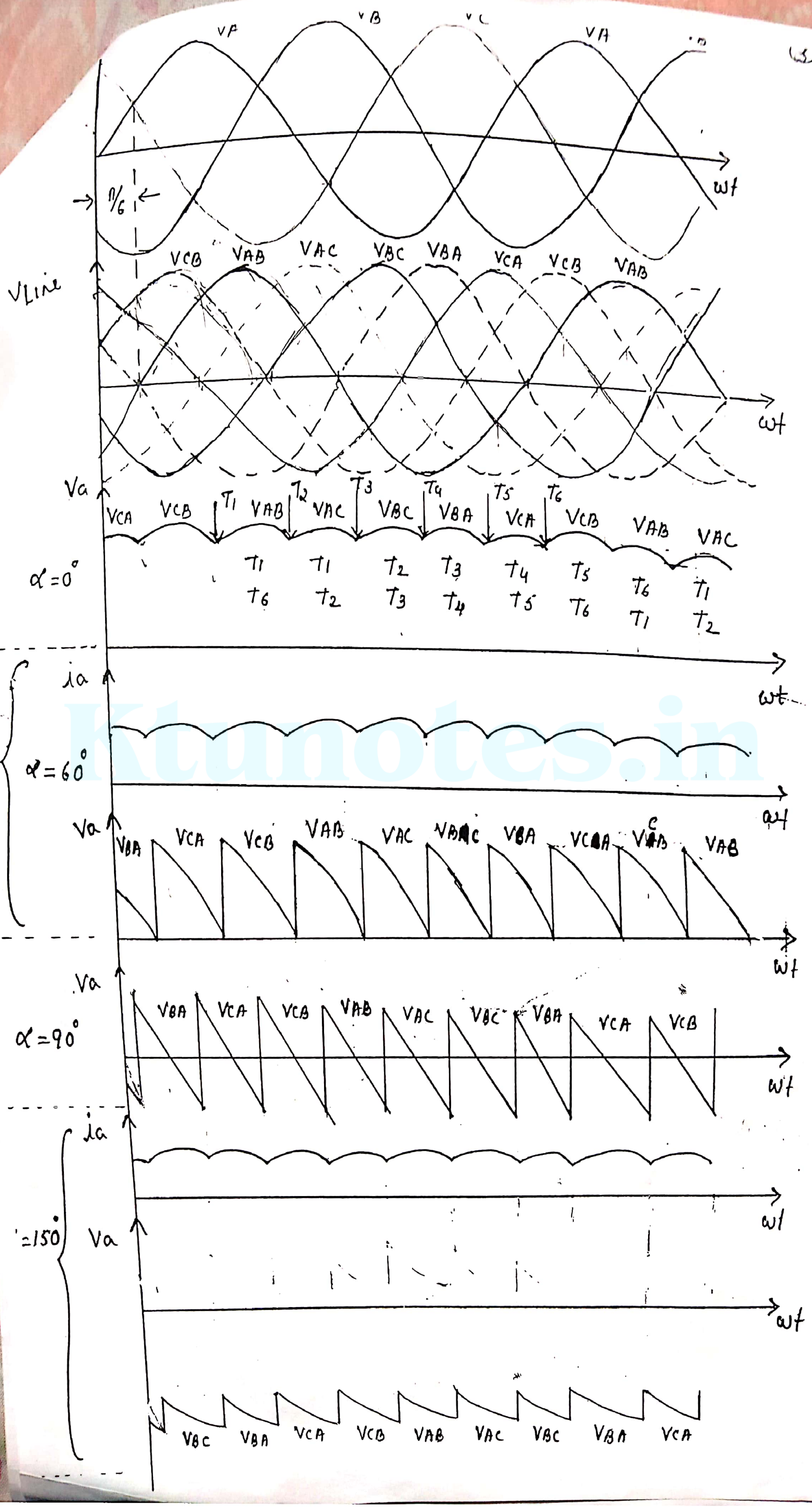
The output voltage can be drawn by considering the line voltages and the phase voltages. The line voltage voltages will be shifted $\pi/6$ by right.

when $\alpha = 60°$

T_1 triggered at $\omega t = 90°$ (30+60)

T_2 triggered at $\omega t = 150$ (90+60)

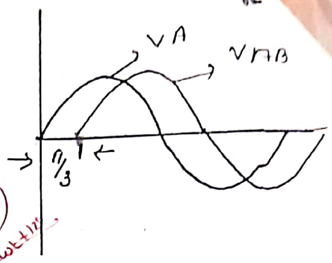
considering the line voltages T_1 is trigged at $\omega t = 120°$ & T_2 at $\omega t =$ 180°


VA
VB
VC
VA
ωt
π/6
VLine
VCB
VAB
VAC
VBC
VBA
VCA
VCB
VAB
ωt
Va
VCA
VCB
T1
VAB
T2
VAC
T3
VBC
T4
VBA
T5
VCA
T6
VCB
VAB
VAC
α = 0°
T1 T6
T1 T2
T2 T3
T3 T4
T4 T5
T5 T6
T6 T1
T1 T2
ia
ωt
α = 60°
ωt
Va
VBA
VCA
VCB
VAB
VAC
VBC
VBA
VCA
VCB
VAB
ωt
Va
α = 90°
VBA
VCA
VCB
VAB
VAC
VBC
VBA
VCA
VCB
ωt
ia
ωt
=150°
Va
ωt
VBC
VBA
VCA
VCB
VAB
VAC
VBC
VBA
VCA

$0 < \alpha < 60$

Average output voltage α

$$V_{a(avg)} = \frac{3}{\pi}\int_{\pi/6+\alpha}^{\pi/2+\alpha} (V_A - V_B)\, d\omega t$$



$V_A = V_m \sin\omega t$
$V_B = V_m \sin(\omega t - 120)$
$= V_m \sin \omega t - \ldots$
$V_m \sin \omega t - V_1$

$V_A - V_B = V_{AB}$

$$V_{AB} = \sqrt{3}\, V_m \sin(\omega t + \pi/6)$$

$\boxed{V_L = \sqrt{3}\, V_{ph}}$

Similarly

$$V_{BC} = V_B - V_C = \sqrt{3}\, V_m \sin(\omega t - \pi/2)$$

$$V_{CA} = V_C - V_A = \sqrt{3}\, V_m \sin(\omega t + \pi/2)$$

$V_A = V_m \sin \omega t$

$$V_{a(avg)} = \frac{3}{\pi}\int_{\pi/6+\alpha}^{\pi/2+\alpha} \sqrt{3}\, V_m \sin(\omega t + \pi/6)$$

$$= \frac{\sqrt{3}\, V_m\, 3}{\pi}\Big[-\cos(\omega t + \pi/6)\Big]_{\pi/6+\alpha}^{\pi/2+\alpha}$$

$$= \frac{3\sqrt{3}\, V_m}{\pi} \times \Big[-\cos(\pi/2 + \alpha + \pi/6) + \cos(\pi/6 + \alpha + \pi/6)\Big]$$

$$= \frac{3\sqrt{3}\, V_m}{\pi}\Big[\cos(\pi/3 + \alpha) - \cos(2\pi/3 + \alpha)\Big]$$

$\cos a - \cos b$
$\sin\frac{(a+b)}{2} \cdot \sin\frac{(a-b)}{2}$

$$= \frac{3\sqrt{3}\, V_m}{\pi} \times 2\left[\sin\left(\frac{\pi/3 + \alpha + 2\pi/3 + \alpha}{2}\right) \sin\left(2\pi/3 + \alpha - \pi/3 - \alpha\right)\right]$$

$$= \frac{3\sqrt{3}\, V_m}{\pi} \times 2\Big[\sin(\pi/2 + \alpha) \cdot \sin \pi/6\Big]$$

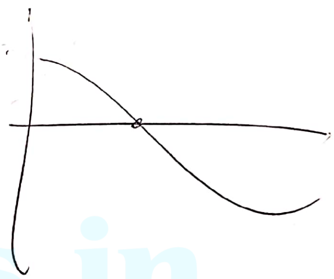
$$V_a(avg) = \frac{3\sqrt{3}V_m}{\pi} \times 2 \times \cos\alpha \times \frac{1}{2}$$

$$\boxed{V_a(avg) = \frac{3\sqrt{3}\,V_m \cos\alpha}{\pi}}$$

$$\boxed{\begin{aligned} &\cos A - \cos B = 2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{B-A}{2}\right). \\ &\sin(90+\theta) = \cos\theta \end{aligned}}$$

Maximum value of the average value will be at $\alpha = 0°$

$$V_a(avg)_{max} = \frac{3\sqrt{3}\,V_m}{\pi}$$



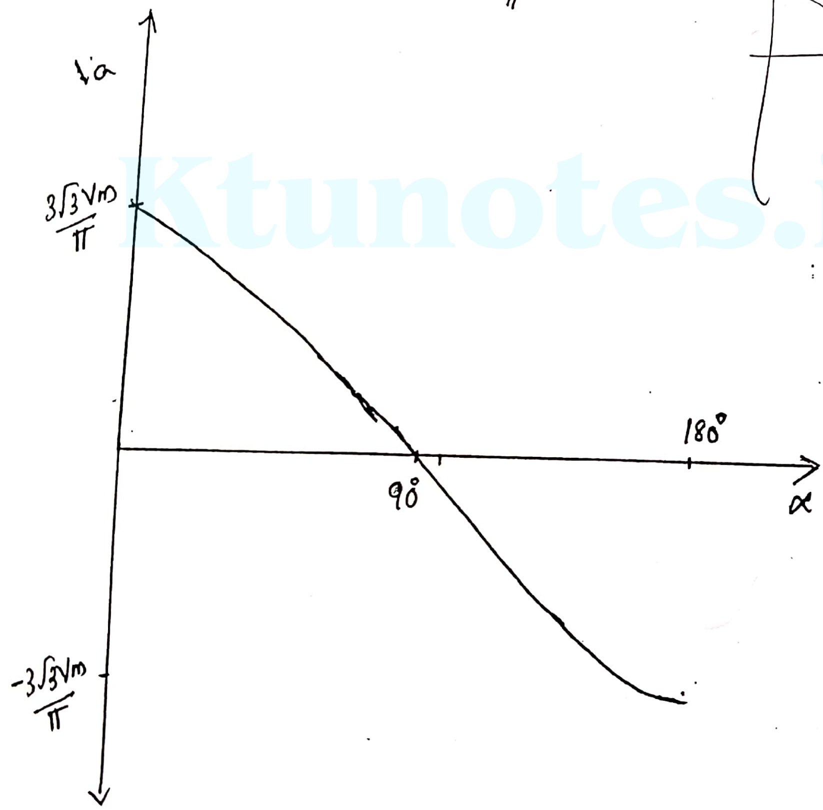


Fig shows the variation of Average voltage Va with firing angle α.

while writing the motor equation

$$V_a(avg) = I_a R_a + E$$

$E = K_b \omega_m$

$T = K_a I_a$

$K_a = K_b = K$

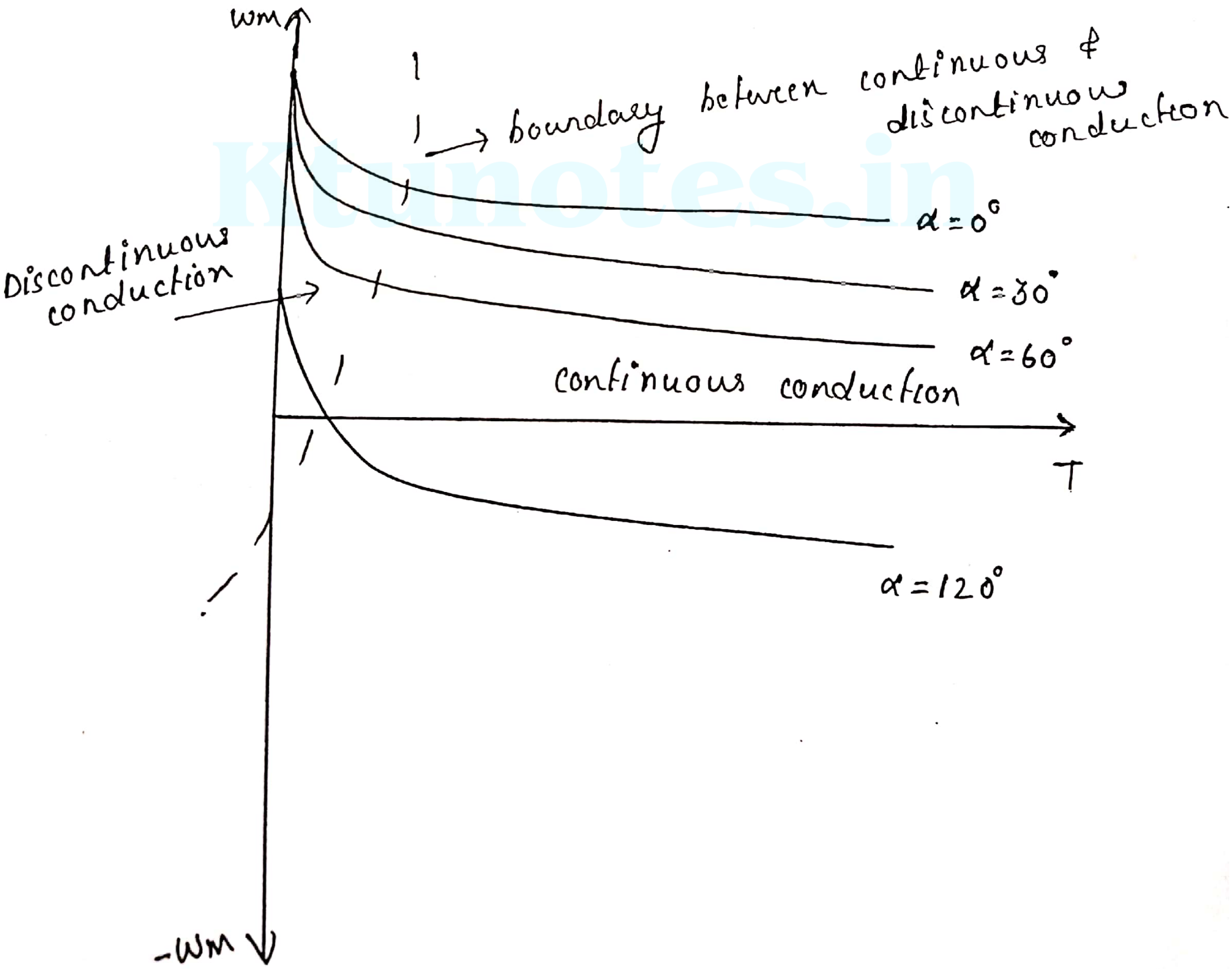
$$\frac{3\sqrt{3}V_m}{\pi}\cos\alpha = \frac{TR_a}{K_b} + K_b\omega_m$$

where $\omega_m \rightarrow$ motor speed in rad/sec.

$$\frac{3\sqrt{3}V_m}{\pi}\cos\alpha - \frac{TR_a}{K_b} = K_b\omega_m$$

$$\boxed{\omega_m = \frac{3\sqrt{3}V_m}{\pi K_b}\cos\alpha - \frac{TR_a}{K_b^2}.}$$

Speed Torque characteristics



Problem

A 150 HP, 500V separately excited dc motor running at ~~1100~~ 1650 rpm is controlled by a 3φ full converter, which is operating from a 3φ, 460V, 50Hz supply. The rated armature current of a motor is 150A. Given motor parameters are $R_a = 0.099\Omega$, $L_a = 0.63mH$. and $K_a\phi = 0.33 V/rpm$ $\neq K_b$

The losses in the system are neglected. Find

a) At $\alpha = 0°$ and $\alpha = 45°$, no-load speeds, assuming that at no load the armature current is 10% of the rated current and is continuous. Hence find

b) Firing angle at 1650 rpm at rated motor current. Compute supply power factor and supply power.

c) For firing angle obtained in (b) find speed regulation

Answer

a) supply phase voltage $= \frac{460}{\sqrt{3}} = 265.58V$

$$V_a = \frac{3\sqrt{3}V_m \cos\alpha}{\pi}$$

$$= \frac{3\sqrt{3} \times \sqrt{2}}{\pi} \times \frac{460}{\sqrt{3}} \cos\alpha$$

$$V_a = 621.22 \cos\alpha$$

$$\alpha = 0°, \quad V_a = \underline{\underline{621.22V}}$$

$$E_1 = V - I_{a0}R_a = 621.22 - 15 \times 0.099$$

$$= \cancel{606.37V} \quad \underline{\underline{619.735V}}$$

no load current is 10% of the rated current

speed for $E_1 = 619.735V$

$$N_1 = \frac{E}{K_a\phi} = \frac{619.735}{0.33} = \underline{\underline{1877.98 rpm}}$$

for $\alpha = 45^\circ$

$$V_a = 621.22 \cos 45$$
$$= \underline{\underline{439.06\ V}}$$

$$E_\alpha = V_a - I_{ao} R_a$$
$$= 439.06 - 15 \times 0.099$$
$$= \underline{\underline{437.575 V}}$$

speed $N_2 = \dfrac{437.575}{0.33} = 1325.9848\ rpm.$

b) Full load condition

$$E = 0.33 \times 1650 = \underline{\underline{544.5 V}}$$

$$V_a = E + I_a R_a$$
$$= 544.5 + 150 \times 0.099$$
$$= \underline{\underline{559.35\ V}}$$

$$559.35 = 621.22 \cos \alpha$$

$$\cos \alpha = \frac{559.35}{621.22}$$

$$\alpha = \underline{\underline{25.78^\circ}}$$

The rms input current

$$I_{rms} = \left[\frac{1}{\pi} \int_{\pi/6+\alpha}^{5\pi/6+\alpha} I_a^2 \, d\omega t \right]^{1/2}$$

$$= \left[\frac{1}{\pi} I_a^2 \left(\omega t \right)_{\pi/6+\alpha}^{5\pi/6+\alpha} \right]$$

$$\left[\frac{Ia^2}{\pi}\left(\frac{5\pi}{6} + \alpha - \frac{\pi}{6} - \alpha \right) \right]$$

$$= \left[\frac{Ia^2}{\pi}\left(\frac{4\pi}{6} \right) \right]^{1/2}$$

$$I_{rms} = \left[Ia^2 \times \frac{2}{3} \right]^{1/2} = \sqrt{2/3}\, Ia.$$

$$I_{rms} = \sqrt{2/3} \times 150$$

$$= \underline{\underline{122.47A}}$$

Supply power $VA = 3 V I_{rms}$

$$= 3 \times \frac{460}{\sqrt{3}} \times 122.47$$

$$= \underline{\underline{97580.7358}} \text{ Watts}$$

Power output $= Va\, Ia$

$$= 559.39 \times 150$$

$$= \underline{\underline{83908.5}} \text{ watts}$$

$$\text{supply power factor} = \frac{\text{o/p power}}{\text{i/p power}}$$

$$= \frac{83908.5}{97580.7358}$$

$$= \underline{\underline{0.859}} \text{ lag}$$

c) speed regulation

Full load current $= 150A$

10% of full load current $= 15A$.

E at 10% of I_a

$$E = 559.35 - 15 \times 0.099$$
$$= 557.865 V$$

No load speed $= \dfrac{557.865}{0.33} = 1690.5$ rpm

speed regulation $= \dfrac{1690.5 - 1650}{1650} \times 100$

$= 2.45\%$